

MODEL QUESTION PAPER 1**Transportation Engineering***Time : 3 Hour**Max.Marks : 75***PART A****I. Answer all questions in one word or one sentence (9 x 1 = 9 Marks)**

1. Indian Road Congress (I.R.C.) was founded as per the suggestion of -----
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2. The position or the layout of the centre line of the highway on the ground is called -----
3. ----- is also called design gradient.
4. ----- layer provides structural support for the pavement surface.
5. The clear minimum perpendicular distance between the inner faces of the two rails is called
6. A mechanical installation which enabling trains to be guided from one track to another
7. The end supports of the superstructure of the bridge is called-----
8. ----- are the buildings used for servicing and repairs of the aircraft
9. The solid parallel platform in a harbour with berthing facility on one side only is called

PART B**II. Answer any eight questions****(8 x 3 = 24 Marks)**

1. List the IRC classification of roads
2. Illustrate turbine junction in highway with the help of a neat sketch
3. Explain super elevation
4. List the types of curves used in highway
5. State the desirable properties of bitumen.
6. Explain the types of railway guage
7. List out the requirements of an ideal rail
8. Explain the necessity of tunnels
9. State the purposes of breakwaters
10. Name the various parts of a bridge with its function

PART C

Answer ALL questions. Each question carries 7 marks (6 x 7 = 42Marks)

- III Explain various traffic studies that can be conducted to analyse traffic characteristics

OR

- IV Explain the various types of sign boards that you have noticed in highways
V Explain the construction procedure for cement concrete roads

OR

- VI Explain the various types of gradients used in highway
VII Explain the various types of sight distances

OR

- VIII Explain the structural layers of pavement with their functions
IX Illustrate the cross section of railway in embankment

OR

- X Explain classification of station yards
XI Illustrate a right hand turn out with the help of a neat sketch

OR

- XII Explain the importance of interlocking of signals
XIII Illustrate the various configurations of runways

OR

- XIV Draw the layout of a harbour and mark all the component parts